

AN EXPLICIT SEPARATION OF THE TYPE AND TEST MODEL STRUCTURES ON DE MORGAN CUBICAL SETS

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ABSTRACT. On presheaves over the De Morgan cube category, the weak equivalences $\mathcal{W}_{\text{type}}$ of the cubical-type-theoretic model structure and the weak equivalences $\mathcal{W}_{\text{test}}$ of the test structure satisfy $\mathcal{W}_{\text{type}} \subseteq \mathcal{W}_{\text{test}}$. Recent published work records the expected non-coincidence, following a 2018 folklore argument of Sattler and Coquand. Here we give an explicit direct separation: a concrete map $\bar{\Phi} \in \mathcal{W}_{\text{test}} \setminus \mathcal{W}_{\text{type}}$, from a Klein quotient of the join $\square^1 \star \square^1$ to the Klein quotient of the square, built from the De Morgan median interpolation. On the test side both objects realize $B\mathbb{Z}/2 \star B\mathbb{Z}/2$ and $\bar{\Phi}$ is an equivalence. On the type side we prove a nullity theorem — every map from $\square^2/K$ into a fibrant replacement of the join model is null-homotopic — via a strictification lemma, two three-line parity arguments in free De Morgan algebras, and an equivariant slide along uniform Kan filling cylinders.

1. INTRODUCTION

Let $\square_{\text{DM}}$ be the De Morgan cube category: objects $[n]$, morphisms $[m] \rightarrow [n]$ the n -tuples of free De Morgan polynomials in m variables. This is the cube category with faces, degeneracies, symmetries, diagonals, connections and reversals [9, 2], the site on which CCHM cubical type theory [5] is modelled. The presheaf category $\widehat{\square_{\text{DM}}}$ carries two homotopy theories:

- the *type structure*: the model structure underlying the interpretation of cubical type theory [5, 11], with cofibrations the monomorphisms and fibrations the uniform Kan fibrations; write $\mathcal{W}_{\text{type}}$ for its weak equivalences;
- the *test structure*: $\square_{\text{DM}}$ is a test category in the sense of Grothendieck [6, 10], so $\widehat{\square_{\text{DM}}}$ models the homotopy theory of spaces, a map being a weak equivalence ($\mathcal{W}_{\text{test}}$) exactly when the nerve of its category of elements is a weak homotopy equivalence.

The companion paper [14] proved the comparison

$$(U) \quad \mathcal{W}_{\text{type}} \subseteq \mathcal{W}_{\text{test}}$$

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(on any totally aspherical cube category with this structure). That the inclusion should be strict has been expected since the June 2018 argument of Sattler and Coquand ([12]; see the account in [14]), which indicated that cubical models with reversal do not present spaces; the recent published landmark [3] asserts the expectation in its introduction, citing the 2018 lecture, and [14] completed the key computation of that argument (the interval quotient by reversal is a $K(\mathbb{Z}/2, 1)$ in both structures). The folklore route, however, passes through a comparison with spaces whose precise functorial content is delicate on this site (the geometric realization is *not* a Quillen equivalence for the test structure of $\square_{\text{DM}}$, because of the twisted degeneracies introduced by reversal), and [14] showed that over the square no isotropy quotient separates the two structures by the contractibility criterion, so no *object-level* witness of the expected kind exists there.

This paper gives an explicit direct separation: a concrete map in $W_{\text{test}} \setminus W_{\text{type}}$, together with the finite rigidity mechanism explaining the failure.

Theorem 1.1 (Main theorem). *Let $K \cong \mathbb{Z}/2 \times \mathbb{Z}/2$ act on $\square^2$ by the coordinate swap and the double reversal, and on the presheaf join $J := \square^1 \star \square^1$ by the coordinatewise reversals; let $W := \square^2/K$ be the strict Klein quotient. The equivariant median collage*

$$\bar{\Phi} : J/K \longrightarrow W$$

of Section 3 satisfies $\bar{\Phi} \in W_{\text{test}} \setminus W_{\text{type}}$. Consequently $W_{\text{type}} \subsetneq W_{\text{test}}$ on $\widehat{\square_{\text{DM}}}$.

The two halves of the proof are of independent interest.

The test side. The strict quotient W is a small object with a large test homotopy type: by [14], $\text{el}(W) \simeq B\mathbb{Z}/2 \star B\mathbb{Z}/2 \simeq \mathbb{RP}^\infty \star \mathbb{RP}^\infty$, a simply connected space with $H_3 = \mathbb{Z}/2$. We construct a comparison object on the join side: the K -quotient of the presheaf join $\square^1 \star \square^1$, whose two ends are the interval quotients $L = \square^1/\langle \neg \rangle$ studied in [14], and a map $\bar{\Phi} : J/K \rightarrow W$ assembled from the *De Morgan median interpolation*

$$F(d, a, t) = (d \wedge \neg t) \vee (a \wedge t) \vee (d \wedge a),$$

whose equivariance is exactly the classical self-duality of the median (Lemma 3.2). Theorem 4.1 proves that $\text{el}(\bar{\Phi})$ is a weak homotopy equivalence. The proof pulls the sieve-cosieve decomposition of $\text{el}(W)$ back along the comparison and checks the three legs of the collage; the required deformation retractions are produced by a *flow identity* for F (Lemma 3.3), an absorption computation in the free De Morgan algebra.

The type side. Here the strictness of the quotient bites. Maps out of a strict quotient are classified by a *strictification lemma* (Lemma 5.1): a map $W \rightarrow Y$ is the same thing as a single square of Y strictly invariant under the K -action on its two coordinates. Theorem 6.3 (the nullity theorem) then shows that *every* map from W into a fibrant replacement R of J/K in the type structure is null-homotopic. The engine is a pair of three-line

parity arguments in the free De Morgan algebra (Lemmas 5.2 and 5.3): strict invariance can never be achieved with a genuine sign twist, so every invariant cell of J/K admits a strictly invariant representative and factors through a contractible cover or a null end; and every invariant *filler* cell of the replacement — which uniformity confines to representable-shaped filler values (Lemmas 2.1 and 2.2) — slides down its own filling cylinder, K -invariantly, onto earlier material. Every case is closed by an explicit one-line null-homotopy. Since W is not contractible in the type structure (a consequence of (U) and the test-side computation), a type-inverse to $\bar{\Phi}$ would exhibit the identity of W as null — a contradiction.

What the separation means. It has been known since [14] (completing a 2018 folklore claim) that neither structure presents the homotopy type of the geometric realization on strict quotients: the interval quotient L has contractible realization but is a $K(\mathbb{Z}/2, 1)$ in *both* structures. On all previously examined strict quotients the two structures agreed with *each other*, and [14] proved an isotropy-agreement statement covering the fixed-point-full and free families. The present theorem shows the agreement fails for the mixed family: on the Klein quotient, the test structure sees the join $\mathbb{RP}^\infty \star \mathbb{RP}^\infty$, while in the type structure there is not even an essential map to (the type realization of) that join. The homotopy theory presented by cubical type theory on the De Morgan site is strictly finer than the test homotopy theory: definitional identifications that the test structure absorbs into a homotopy colimit are, in the type structure, obstructed by the rigidity of strict equivariance — a rigidity we isolate in two parity lemmas.

Honest positioning. The separation is not produced by a new machine; it is produced by locating the exact finite obstruction. The positive ingredients — the collage computation of $\text{el}(W)$, the inclusion (U), the non-contractibility of W — are from [14]. The new ingredients are: the equivariant median collage $\bar{\Phi}$ and its test-side equivalence; the strictification lemma; the two parity lemmas; and the nullity theorem, whose case analysis is elementary once the parity lemmas are in place. All finite combinatorial claims (the parity lemmas, the classification of invariant cells, the equivariance and flow identities) are additionally verified by exhaustive machine computation in the artifact accompanying the series; the proofs given here are complete without the machine.

Outline. Section 2 fixes the two structures and recalls what is imported from [14]. Section 3 constructs J/K and $\bar{\Phi}$. Section 4 proves the test-side equivalence. Section 5 proves the strictification lemma and the parity lemmas, and classifies the invariant cells. Section 6 proves the nullity theorem. Section 7 assembles the separation and its corollaries. Section 8 describes the machine cross-checks.

2. PRELIMINARIES

2.1. The De Morgan cube category and its presheaves. We write $\text{DM}(i_1, \dots, i_m)$ for the free De Morgan algebra on generators $i_1, \dots, i_m$: the free bounded distributive lattice with an involution $\neg$ satisfying the De Morgan laws, *without* excluded middle. Its elements have canonical disjunctive normal forms (antichains of clauses over the literals) [5]. $\square_{\text{DM}}$ has objects $[m]$, $m \geq 0$, and $\square_{\text{DM}}([m], [n]) := \text{DM}(i_1, \dots, i_m)^n$, with composition by substitution. We write $\widehat{\square}_{\text{DM}}$ for the presheaf category, $\square^n$ for the representable on $[n]$, and use x, y (resp. d, a, t) for the coordinates of $\square^2$ (resp. $\square^3$).

We will use the representation of the free algebra by monotone functions: DM_n is isomorphic to the lattice of monotone maps $\{0, 1\}^{2n} \rightarrow \{0, 1\}$ on the *literal cube* L_n , with coordinates in pairs $(x_i, \neg x_i)$; the involution is

$$(1) \quad (\neg\varphi)(p) = 1 - \varphi(\rho p),$$

where ρ swaps each literal pair and complements it [13]. Two total involutions of the literal cube will appear below: the *generator swap* sw (exchange the x -pair with the y -pair) and the *literal swap* composed with ρ , which is the bitwise complement $c(p) = \bar{p}$.

2.2. The two structures. The *type structure* on $\widehat{\square}_{\text{DM}}$ is the model structure underlying the CCHM interpretation: cofibrations are the monomorphisms, fibrations the *uniform* Kan fibrations — maps equipped with fillers for the open boxes $(m \hat{\otimes} \delta^\varepsilon)$, m a monomorphism, chosen strictly naturally in pullback along substitutions — and weak equivalences $\mathcal{W}_{\text{type}}$; see [5] for the type theory, [7] for uniform fibrations, and [11] for the model structure. Every object is cofibrant.

Two structural facts about fibrant replacement are used; both hinge on *uniformity*. Recall that in any presheaf category the monomorphisms are cellularly generated by a set of inclusions with quotients of representables as codomains (existence of cellular models, Cisinski [6, §1.2]); on a site with diagonals, connections and reversals — which is not (generalized) Reedy, cf. the discussion in [4] — the quotient shapes are genuinely needed, and representable boundaries alone do not suffice. Uniformity is what nevertheless confines fibrant replacement to representable-shaped data.

Lemma 2.1 (Descent via uniformity). *An object equipped with fillers for the boxes $(S \hookrightarrow \square^n) \hat{\otimes} \delta^\varepsilon$, where S ranges over arbitrary subpresheaves of representables, strictly natural in substitution, is fibrant.*

Proof sketch. A lifting problem against $m \hat{\otimes} \delta^\varepsilon$ for an arbitrary monomorphism m reduces, attaching one nondegenerate cell orbit at a time, to boxes over cell shapes $Q = \square^n/E$ relative to their proper subobjects (on this non-Reedy site the cell shapes are genuinely quotients [4], which is why representable *boundaries* alone would not suffice). Such a box pulls back along the projection $\square^n \rightarrow Q$ to a box of the stated form, with S the preimage

subpresheaf; the chosen filler is strictly natural under all substitutions, in particular under the pairs identified by the congruence E , so it is constant on E -classes and descends to a filler over Q . Compatibility with the box and with earlier attachments is again naturality. $\square$

Let $\mathbb{J}$ denote the small category whose objects are the box inclusions $j_{S,n,\varepsilon} := (S \hookrightarrow \square^n) \hat{\otimes} \delta^\varepsilon$ of Lemma 2.1, and whose morphisms $j_{S',m,\varepsilon} \rightarrow j_{S,n,\varepsilon}$ are the substitutions $\sigma \in \square_{\text{DM}}([m], [n])$ carrying S' into S ; such a σ induces the box morphism $\hat{\sigma} := \sigma \times \text{id}_{\square^1}$. A *box* for $j = j_{S,n,\varepsilon}$ over X is a map $b : \text{dom}(j) = S \times \square^1 \cup \square^n \times \{\varepsilon\} \rightarrow X$; a morphism σ restricts boxes, $b \mapsto b \circ \hat{\sigma}$. Uniform fibrancy is precisely a right lifting structure against the *category* $\mathbb{J}$ (fillers for all boxes, compatible with restriction), the formulation of [7].

Lemma 2.2 (Algebraic fibrant replacement). *Every $X \in \widehat{\square_{\text{DM}}}$ admits a functorial trivial cofibration $u : X \rightarrow R$ with R uniformly fibrant, constructed as the colimit of a transfinite sequence $X = R_0 \rightarrow R_1 \rightarrow \dots$ of monomorphisms in which*

$$R_{\beta+1} := \text{coeq} \left(\coprod_{\substack{\sigma: j' \rightarrow j \\ b \text{ box for } j \text{ over } R_\beta}} \text{cod}(j') \rightrightarrows \coprod_{(j,b)} \text{cod}(j) \right) \sqcup_{\coprod_{(j,b)} \text{dom}(j)} R_\beta,$$

the two parallel arrows being the insertion at $(j', b \circ \hat{\sigma})$ and the composite of $\hat{\sigma}$ with the insertion at (j, b) , and the pushout taken along the evaluations b . Consequently:

- (1) *every cell of $R_{\beta+1}$ is a cell of R_β or a filler value $\text{val}(b, c)$, $c \in \text{cod}(j)$, and the identifications among filler values are exactly those generated by*
 - (i) $c \in \text{dom}(j)$: $\text{val}(b, c) = b(c)$, a cell of R_β (the pushout);
 - (ii) $\text{val}(b \circ \hat{\sigma}, c) = \text{val}(b, \hat{\sigma} \circ c)$ (the coequalizer);
- (2) *the stage maps are monomorphisms and no identifications are imposed on old cells: the evaluations agree on the cells identified by the coequalizer, precisely because $b \circ \hat{\sigma}$ is by definition the restricted box; hence equality of cells of R is detected at a stage and is generated by (i)–(ii);*
- (3) *filler values are strictly natural, $\text{val}(b, c) \cdot f = \text{val}(b, c \cdot f)$, and R carries the tautological uniform filling structure, so R is fibrant by Lemma 2.1;*
- (4) *by the standard property of the algebraically free awfs generated by $\mathbb{J}$, the unit u lies in the left class cofibrantly generated by the box inclusions — Garner’s construction interposes coequalizers, so u is a retract of a transfinite composite of pushouts of the generators rather than literally such a composite — and since the box inclusions are trivial cofibrations of the type structure, whose left class is closed under pushout, transfinite composition and retract, u is a trivial cofibration; the construction is functorial in X .*

Proof. This is Garner’s algebraic small object argument [8] for the generating *category* $\mathbb{J}$ (rather than a mere set), in the free-monad-on-a-pointed-endofunctor form; the displayed one-step construction is its explicit colimit presentation, as in the treatment of uniform fibrations in [7]. The listed properties are read off from the presentation: (1) is the description of a coequalizer-then-pushout of coproducts of presheaves, whose cell-level relations are precisely the displayed generators; for (2), both relation schemes have a fresh member on each side except (i), whose old member is uniquely determined, and chains through fresh cells terminating in old cells on both ends force equal old cells because restricted boxes evaluate compatibly ($b(\hat{\sigma} \circ c) = (b \circ \hat{\sigma})(c)$ on the box domain); (3) the presheaf structure on a coproduct of representable-products acts on the c -coordinate, and the filling structure assigns to a box its inserted values, uniformity holding by (ii); (4) is the standard comparison of the algebraically free awfs with the cofibrantly generated one [8]. $\square$

The *test structure*: $\square_{\text{DM}}$ is a strict test category (the cube category with connections is one by [10]; the De Morgan case is covered by the criteria of [6], cf. [2]), so the functor $X \mapsto N(\text{el}(X))$ detects a class $\mathcal{W}_{\text{test}}$ of weak equivalences making $\widehat{\square_{\text{DM}}}$ a model for the homotopy theory of spaces. We freely write $\text{el}(X) \simeq Z$ for “the nerve of the category of elements of X is weakly homotopy equivalent to Z ”.

From [14] we import:

- (U) $\mathcal{W}_{\text{type}} \subseteq \mathcal{W}_{\text{test}}$.
- (W2) Let $K = \{1, \text{sw}, \text{nb}, g\}$ act on $\square^2$ by $\text{sw}(x, y) = (y, x)$, $\text{nb}(x, y) = (\neg x, \neg y)$, $g = \text{nb} \circ \text{sw}$, and let $W := \square^2/K$ (levelwise orbits). Then $\text{el}(W) \simeq B\mathbb{Z}/2 \star B\mathbb{Z}/2$: the category of elements decomposes into the stabilized sieve (the diagonal and antidiagonal layers, each $\simeq B\mathbb{Z}/2$) and the free cosieve ($\simeq BK$), and the collage evaluates to the join. In particular $\text{el}(W)$ is simply connected with $H_3(\text{el}W; \mathbb{Z}) = \mathbb{Z}/2$; it is not weakly contractible.
- (L) $L := \square^1/\langle \neg \rangle$ satisfies $\text{el}(L) \simeq B\mathbb{Z}/2$; the reversal acts freely on the nonempty cells of $\square^1$ (no element of a free De Morgan algebra equals its own negation), and L is a $K(\mathbb{Z}/2, 1)$ in both structures.
- (Collage) For a sieve–cosieve decomposition of a small category, the nerve is the homotopy colimit of the two parts over the two-sided category of elements of the connecting profunctor (simplicial replacement [1] plus [16]); the formula is natural in functors that preserve the sieve.

Corollary 2.3 (W1). *W is not contractible in the type structure: the map $W \rightarrow 1$ is not in $\mathcal{W}_{\text{type}}$.*

Proof. By (U) it would be in $\mathcal{W}_{\text{test}}$, contradicting (W2). $\square$

2.3. Quotients and strictness. Throughout, quotients X/G are *strict*: levelwise orbit sets, the colimit of the group action in $\widehat{\square_{\text{DM}}}$. The tension

studied in this paper is precisely between the strict quotient and the homotopy quotient: the test structure computes the latter from the former (el turns the strict orbit category into the Borel construction, as in (W2)), while the type structure, as we shall prove, does not.

3. THE JOIN MODEL AND THE EQUIVARIANT MEDIAN COLLAGE

3.1. The join and its K -action. Let $J := \square_d^1 \star \square_a^1$ denote the presheaf join: the double mapping cylinder

$$J = \square_d^1 \sqcup_{(\square_d^1 \times \square_a^1) \times \{0\}} (\square_d^1 \times \square_a^1) \times \square_t^1 \sqcup_{(\square_d^1 \times \square_a^1) \times \{1\}} \square_a^1,$$

concretely: cells at level m are triples $(\delta, \alpha, \tau) \in \text{DM}(i_1 \dots i_m)^3$, modulo the end collapses ($\tau = 0$: the α -component is forgotten; $\tau = 1$: the δ -component is forgotten). K acts by

$$\text{sw} \cdot (\delta, \alpha, \tau) = (\delta, \neg\alpha, \tau), \quad \text{nb} \cdot (\delta, \alpha, \tau) = (\neg\delta, \neg\alpha, \tau),$$

so $g = (\neg\delta, \alpha, \tau)$; the action is by reversals of the two join coordinates, trivial on the cylinder coordinate. The $t=0$ end of J/K is $\square_d^1$ modulo the residual reversal, i.e. a copy of L (the *diagonal end* L_D); the $t=1$ end is another copy (L_A); on the open cylinder the action is free (no self-negated elements).

3.2. The median interpolation.

Definition 3.1. $F(d, a, t) := (d \wedge \neg t) \vee (a \wedge t) \vee (d \wedge a) \in \text{DM}(d, a, t)$.

F interpolates: $F(d, a, 0) = d$ and $F(d, a, 1) = a$, by absorption. Its two key properties:

Lemma 3.2 (Self-duality). $\neg F(d, a, t) = F(\neg d, \neg a, t)$.

Proof. $\neg F = (\neg d \vee t) \wedge (\neg a \vee \neg t) \wedge (\neg d \vee \neg a)$. Distributing the first two factors gives $(\neg d \wedge \neg a) \vee (\neg d \wedge \neg t) \vee (t \wedge \neg a) \vee (t \wedge \neg t)$; meeting with $(\neg d \vee \neg a)$ and absorbing (each of the last three joinands is below $\neg d$ or $\neg a$ up to the terms retained) yields $(\neg d \wedge \neg t) \vee (\neg a \wedge t) \vee (\neg d \wedge \neg a) = F(\neg d, \neg a, t)$. $\square$

Lemma 3.3 (Flow identity). $F(d, a, t \wedge s) = F(d, F(d, a, t), s)$ and dually $F(d, a, t \vee s) = F(F(d, a, t), a, s)$.

Proof. Expand the right side with $F := F(d, a, t)$: $(d \wedge \neg s) \vee (F \wedge s) \vee (d \wedge F) = (d \wedge \neg s) \vee (d \wedge \neg t \wedge s) \vee (a \wedge t \wedge s) \vee (d \wedge a \wedge s) \vee (d \wedge \neg t) \vee (d \wedge a \wedge t) \vee (d \wedge a)$. Absorption removes the three joinands below $d \wedge \neg t$ or $d \wedge a$, leaving $(d \wedge (\neg s \vee \neg t)) \vee (a \wedge t \wedge s) \vee (d \wedge a) = F(d, a, t \wedge s)$. The second identity is dual (Lemma 3.2). $\square$

3.3. The collage map.

Definition 3.4. $\Phi := (F(d, a, t), F(d, \neg a, t)) : \square^3 \rightarrow \square^2$.

Proposition 3.5. Φ intertwines the K -actions: $\text{sw} \circ \Phi = \Phi \circ \text{sw}_J$ and $\text{nb} \circ \Phi = \Phi \circ \text{nb}_J$, and Φ respects the end collapses. Hence Φ descends to

$$\bar{\Phi} : J/K \longrightarrow W.$$

Moreover $\bar{\Phi}$ restricts to isomorphisms of the ends onto the stabilized layers: the $t=0$ end L_D maps isomorphically onto the diagonal layer $(\delta \mapsto (\delta, \delta))$, and the $t=1$ end onto the antidiagonal layer $(\alpha \mapsto (\alpha, \neg\alpha))$.

Proof. sw -equivariance is immediate from the definition (the two components swap when $a \mapsto \neg a$). nb -equivariance is Lemma 3.2 applied in each component. At $t = 0$ both components equal d (absorption), so the α -collapse is respected and the end maps to the diagonal; at $t = 1$ the components are $(a, \neg a)$. The residual reversals match on both sides, giving isomorphisms of the end layers. $\square$

Geometrically, $\bar{\Phi}$ reads a point of the Klein-quotient square in *join coordinates*: distance from the diagonal is the cylinder coordinate, and the median convex combination interpolates between the two fixed lines. The map is a presheaf-level realization of the collage that computes $\text{el}(W)$ in (W2).

4. THE TEST SIDE: $\text{el}(\bar{\Phi})$ IS A WEAK EQUIVALENCE

Theorem 4.1. $\text{el}(\bar{\Phi}) : \text{el}(J/K) \rightarrow \text{el}(W)$ is a weak homotopy equivalence. Hence $\bar{\Phi} \in W_{\text{test}}$.

The proof occupies this section. Write $\phi := \text{el}(\bar{\Phi})$. Recall from (W2) the sieve–cosieve decomposition of $\text{el}(W)$: the sieve $S_W =$ the stabilized layers (diagonal $\sqcup$ antidiagonal, with no morphisms between them), the cosieve $U_W =$ the free layer.

4.1. The pulled-back decomposition. Since the preimage of a sieve under a functor is a sieve, set $S'' := \phi^{-1}(S_W)$, $U'' := \phi^{-1}(U_W)$. Sieves of a category of elements correspond to subpresheaves; S'' corresponds to the *collapse locus* $Z = Z_D \sqcup Z_A \subseteq J/K$:

$$Z_D := \{(\delta, \alpha, \tau) : F(\delta, \alpha, \tau) = F(\delta, \neg\alpha, \tau)\}, \quad Z_A := \{\dots = \neg F(\delta, \neg\alpha, \tau)\},$$

equational, hence substitution-closed, conditions. (Z_D contains the $t=0$ end, where both components equal δ ; Z_A contains the $t=1$ end.) By naturality of the collage formula (Collage) in sieve-preserving functors, it suffices to prove that the three legs of

$$\begin{array}{ccccc} NS'' & \longleftarrow & N\hat{W}'' & \longrightarrow & NU'' \\ \downarrow & & \downarrow & & \downarrow \\ NS_W & \longleftarrow & N\hat{W} & \longrightarrow & NU_W \end{array}$$

are weak equivalences.

4.2. Leg one: the collapse locus retracts onto the ends.

Lemma 4.2. Z_D is closed under lowering τ (if $(\delta, \alpha, \tau) \in Z_D$ then $(\delta, \alpha, \tau \wedge s) \in Z_D$ for every s), and dually Z_A under raising τ . Moreover $Z_D \cap \{t=1\} = \emptyset = Z_D \cap Z_A$.

Proof. By the flow identity, $F(\delta, \alpha, \tau \wedge s) = F(\delta, F(\delta, \alpha, \tau), s)$, and likewise with $\neg\alpha$; if the two inner values agree, so do the outer ones. At $t = 1$ the components are α and $\neg\alpha$, never equal (no self-negated elements); membership in both loci would force $F = \neg F$, likewise impossible. $\square$

Consequently the presheaf cylinder $h_s : (\delta, \alpha, \tau) \mapsto (\delta, \alpha, \tau \wedge \neg s)$ restricts to Z_D and deformation-retracts it onto the $t=0$ end, fixing the end pointwise; dually for Z_A . A presheaf homotopy induces, after $Nel(-)$, equality of the maps in the homotopy category (the projection $X \times \square^1 \rightarrow X$ is in $\mathbf{W}_{\text{test}}$). Combining with Proposition 3.5 (the end maps are isomorphisms of layers): $\phi|_{Z_D} \simeq (\text{iso}) \circ (\text{retraction})$, a weak equivalence; the A -component is dual; the two components are disjoint as categories on both sides.

4.3. Legs two and three: middle layers via terminal covers.

Lemma 4.3 (Free quotients of contractible covers). *Let G be a finite group acting on $X \in \widehat{\square}_{\text{DM}}$ such that the action on the nonempty cells is free, and suppose the full subcategory $\mathcal{C} \subseteq \text{el}(X)$ on a G -stable class of cells has contractible nerve. Then: (i) $\text{el}(X/G) \cong \text{el}(X)/G$ as categories, and the corresponding full subcategory of $\text{el}(X/G)$ is the quotient category $\mathcal{C}/G$; (ii) the action of G on $\mathcal{C}$ is free on objects and morphisms, $N(\mathcal{C}/G) \cong N(\mathcal{C})/G$, and the quotient map $N\mathcal{C} \rightarrow N(\mathcal{C}/G)$ is a principal G -covering of simplicial sets; hence $N(\mathcal{C}/G) \simeq BG$; (iii) if $\mathcal{C} \rightarrow \mathcal{D}$ is a G -equivariant functor between two such subcategories (over possibly different G -presheaves), the induced functor $\mathcal{C}/G \rightarrow \mathcal{D}/G$ is a weak equivalence.*

Proof. (i) Cells of X/G are orbits of cells and a morphism $([m], [x]) \rightarrow ([n], [x'])$ is a cube map f with $[x] = [x'] \cdot f$, i.e. $x \cdot g = x' \cdot f$ for a unique g (freeness); this is exactly a morphism of the quotient category. (ii) Freeness on objects gives freeness on morphisms (a morphism fixed by g has fixed source); a free action of a finite group on a small category quotients to a category whose nerve is the quotient simplicial set, and a free simplicial action has quotient a principal covering; since $N\mathcal{C}$ is contractible, the quotient is a $K(G, 1)$. (iii) The equivariant functor induces a map of principal G -coverings over the quotients, hence a π_1 -isomorphism ($= \text{id}_G$ after the deck identifications) between $K(G, 1)$'s, and both quotients are aspherical by (ii); apply Whitehead. $\square$

Lemma 4.4. ϕ restricts to weak equivalences $U'' \rightarrow U_W$ and $\hat{W}'' \rightarrow \hat{W}$.

Proof. U'' is the full subcategory of cells not in Z . Upstairs (before quotienting) the corresponding full subcategory of $\text{el}(\square^3\text{-cylinder})$ contains the generic cell $\text{id}_{[3]}$ (its Φ -components are separated: at (d, a, t) generic the

antichain normal forms differ), and every object maps to it uniquely, so the cover is contractible; K acts freely on the relevant cells (no self-negated elements). The same holds for the free-layer cover of W with terminal object $\text{id}_{[2]}$. Lemma 4.3 now applies verbatim: both quotients are BK 's and the equivariant comparison is a weak equivalence. For $\hat{W}''$: as in [14], the two-sided category is an arrow category fibered over the sieve part with fibers having terminal objects (id, s) , and base contracting (by the flow retraction of Lemma 4.2) onto the end, which is the category of elements of a representable, hence contractible; K acts freely on the arrow component. Both sides are BK and the comparison is again an equivariant map of contractible free covers. $\square$

This completes the proof of Theorem 4.1. (The statement is also consistent with a direct computation: the collage of $\text{el}(J/K)$ evaluates to $\text{hocolim}(B\mathbb{Z}/2 \leftarrow BK \rightarrow B\mathbb{Z}/2) = B\mathbb{Z}/2 \star B\mathbb{Z}/2$, matching (W2).)

5. STRICTIFICATION AND THE PARITY LEMMAS

5.1. Maps out of a strict quotient.

Lemma 5.1 (Strictification). *For every $Y \in \widehat{\square_{\text{DM}}}$, precomposition with the quotient of the Yoneda embedding induces a bijection*

$$\widehat{\square_{\text{DM}}}(W, Y) \cong \{s \in Y([2]) : s \cdot \text{sw} = s, s \cdot \text{nb} = s\},$$

where $k \in K \subseteq \square_{\text{DM}}([2], [2])$ acts by restriction. Likewise $\widehat{\square_{\text{DM}}}(J/K, Y)$ is the set of elements of $Y([3])$ invariant under K_J and compatible with the end collapses, and $\widehat{\square_{\text{DM}}}(W \times \square^1, Y)$ is the set of K -invariant elements of $Y([2+1])$ (invariance in the first two coordinates).

Proof. W is the coequalizer of the K -action on $\square^2$; apply Yoneda. The same for the other two (for the cylinder, $(\square^2 \times \square^1)/(K \times 1) = W \times \square^1$ since quotients are levelwise). $\square$

Thus homotopy-theoretic questions about maps from W become questions about strictly invariant cells — finitely constrained data.

5.2. The parity lemmas. The next two lemmas are the heart of the paper. Recall the representation (1): elements of DM_2 are monotone functions on the literal cube $\{0, 1\}^4$ with coordinates $(v_x, v_{\neg x}, v_y, v_{\neg y})$; negation is $(\neg\varphi)(p) = 1 - \varphi(\rho p)$; the generator swap sw acts by exchanging the pairs; and $\text{lswap} \circ \rho = c$ is the bitwise complement, so that

$$(2) \quad \varphi^{\text{nb}} = \neg\varphi \iff \varphi(cq) = 1 - \varphi(q) \quad \forall q.$$

Lemma 5.2 (First parity lemma). *No $\varphi \in \text{DM}_2$ satisfies $\varphi^{\text{sw}} = \neg\varphi$.*

Proof. The condition reads $\varphi(\text{sw } p) = 1 - \varphi(\rho p)$ for all p . The point $q = (1, 0, 1, 0)$ satisfies $\text{sw } q = q$ and $\rho q = q$; at q the condition becomes $\varphi(q) = 1 - \varphi(q)$. $\square$

Lemma 5.3 (Second parity lemma). *No $\varphi \in \text{DM}_2$ satisfies both $\varphi^{\text{sw}} = \varphi$ and $\varphi^{\text{nb}} = \neg\varphi$.*

Proof. Let $m = (1, 0, 0, 1)$. Its bitwise complement is $cm = (0, 1, 1, 0) = \text{sw } m$. Then symmetry gives $\varphi(cm) = \varphi(m)$ while (2) gives $\varphi(cm) = 1 - \varphi(m)$. $\square$

5.3. Classification of invariant cells. Cells of $(J/K)([2])$ are, per the end normalization: end cells ($\tau \in \{0, 1\}$, one component retained modulo the residual reversal) and cylinder classes $[(\delta, \alpha, \tau)]$, the K_J -orbit $\{(\pm\delta, \pm\alpha, \tau)\}$ (all four sign patterns).

Proposition 5.4 (Rigidity of invariant cells). *Let $[(\delta, \alpha, \tau)]$ be a cylinder class of $(J/K)([2])$ invariant under both coordinate substitutions $\text{sw}, \text{nb} \in \square_{\text{DM}}([2], [2])$. Then δ, α, τ are separately strictly invariant: $\varphi^{\text{sw}} = \varphi = \varphi^{\text{nb}}$ for each component φ .*

Proof. Invariance under sw means $(\delta^{\text{sw}}, \alpha^{\text{sw}}, \tau^{\text{sw}}) = (\pm\delta, \pm\alpha, \tau)$ for one of the four sign patterns. A minus sign on either component is excluded by Lemma 5.2; hence $\delta^{\text{sw}} = \delta$, $\alpha^{\text{sw}} = \alpha$, $\tau^{\text{sw}} = \tau$. Invariance under nb now reads $(\delta^{\text{nb}}, \alpha^{\text{nb}}, \tau^{\text{nb}}) = (\pm\delta, \pm\alpha, \tau)$ with the components already sw -symmetric; a minus sign is excluded by Lemma 5.3. $\square$

Remark 5.5. Proposition 5.4 was found, and is independently confirmed, by an exhaustive machine census: of the 168^3 triples ($|\text{DM}_2| = 168$, the Dedekind number $M(4)$), the census finds 372 invariant classes in total, of which 360 are interior cylinder classes — every one admitting a strictly invariant representative — and the remaining 12 lie on the collapsed ends (Section 8). The same census shows, as context, that no strict section-like phenomena intervene: no invariant class maps under $\bar{\Phi}$ to the generic cell of W .

6. THE NULLITY THEOREM

Fix the algebraic fibrant replacement $u : J/K \rightarrow R$ of Lemma 2.2. Since every object is cofibrant and R is fibrant, homotopy classes $[W, R]$ are strict maps modulo cylinder homotopy, and by Lemma 5.1 both maps and homotopies are strictly invariant cells.

The inductive engine is the following observation. Write $\sim$ for the equivalence relation on index pairs (B, c) generated by the boundary identifications (i) and the naturality reindexings (ii) of Lemma 2.2, so that two filler values coincide exactly when their index data are related; and note that the presheaf action on filler values is $\text{val}(B, c) \cdot f = \text{val}(B, c \cdot f)$. For a box B of shape $(S \hookrightarrow \square^n) \hat{\otimes} \delta^\varepsilon$ and a cell $c = (c_1, \tau)$ of $\square^n \times \square^1$ define the *slide*

$$c_z := (c_1, \tau \wedge \neg z) \quad (\varepsilon = 0), \quad c_z := (c_1, \tau \vee z) \quad (\varepsilon = 1),$$

a cell of $\square^n \times \square^1$ in one further variable z .

Lemma 6.1 (Equivariant filler slide). *The slide is compatible with the relation: $(B, c) \sim (B', c')$ implies $(B, c_z) \sim (B', c'_z)$. Consequently, if $\text{val}(B, c)$ is invariant under a group of substitutions k acting on its level (i.e. $(B, c \cdot k) \sim (B, c)$ for all k), then so is $\text{val}(B, c_z)$ for every z ($(c \cdot k)_z = c_z \cdot k$, as k does not involve z), and the slide is a homotopy from $\text{val}(B, c)$ at $z = 0$ to a cell of the box itself — earlier-stage material — at $z = 1$.*

Proof. It suffices to check the generating relations. (i) *Boundary*: the box domain is $S \times \square^1 \cup \square^n \times \{\varepsilon\}$. If c_1 lies in S then so does the slid cell's first component (it is unchanged); if $\tau = \varepsilon$ then $\tau \wedge \neg z = 0 = \varepsilon$ (resp. $\tau \vee z = 1 = \varepsilon$): the slide preserves the box domain, on which values are the prescribed old cells, and there the relation is equality of old cells, preserved because the slide acts by a substitution. (ii) *Naturality*: the reindexings have the form $(B \cdot \hat{\sigma}, c') \sim (B, \hat{\sigma} \circ c')$ with $\hat{\sigma} = \sigma \times \text{id}_{\square^1}$ a box morphism over a substitution σ of the base shape; since $\hat{\sigma}$ does not touch the cylinder coordinate, $\hat{\sigma} \circ (c'_z) = (\hat{\sigma} \circ c')_z$, and the relation slides. Chains of generators slide term by term. For the last clause: at $z = 1$ the slid cell is $(c_1, 0)$ (resp. $(c_1, 1)$), which lies in $\square^n \times \{\varepsilon\}$, i.e. in the box domain, so the value is a cell prescribed by B over the earlier stage. $\square$

Remark 6.2. Lemma 6.1 deliberately avoids analysing *how* the invariance of a filler value arises. It may arise with $c \cdot k = c$; but it may equally arise through the naturality relation — for instance when the box itself is invariant under an automorphism of its base shape intertwining k — and no coordinatewise rigidity can be inferred. The slide transports the entire relation chain, whatever it is, down the filling cylinder.

Theorem 6.3 (Nullity). *Every map $f : W \rightarrow R$ is homotopic to a constant map.*

Proof. By Lemma 5.1, f corresponds to a K -invariant element $s \in R([2])$. The cells of R are filtered by the stages of the construction of Lemma 2.2; equality of cells is detected at a stage. We induct on the stage at which s appears.

Base of the induction: s is a cell of J/K . Three cases.

- s is a cylinder class. By Proposition 5.4 it has strictly invariant components, so the classifying map $(\delta, \alpha, \tau) : \square^2 \rightarrow \square^3$ is K -invariant on the nose and descends to $\bar{\sigma} : W \rightarrow \square^3$ with f factoring through the representable $\square^3$; the latter contracts to a vertex by the connection homotopy $(u_1, u_2, u_3, z) \mapsto (u_1 \wedge \neg z, u_2 \wedge \neg z, u_3 \wedge \neg z)$, a strict presheaf cylinder, so f is homotopic to a constant.
- s is an end cell, say in L_D . Then f factors through the end inclusion $L_D \hookrightarrow J/K$, which is null-homotopic by the explicit slide $H(\varphi, \zeta) := [(\varphi, 0, \zeta)]$, a presheaf map $L_D \times \square^1 \rightarrow J/K$ with H_0 the inclusion and H_1 constant. Dually for L_A .
- s is a vertex class: f is constant.

Inductive step: s is a fresh filler value. Write $s = \text{val}(B, c)$ as in Lemma 2.2, with $c = (c_1, \tau) \in (\square^n \times \square^1)([2])$. Invariance means $(B, c \cdot k) \sim (B, c)$ for all $k \in K$. Let $H_z := \text{val}(B, c_z)$ be the slide of Lemma 6.1, formed at level $2+1$ (the slide of the level-3 cell $(c_1 \circ \text{pr}, \tau \wedge \neg z)$). By Lemma 6.1 the relation chains witnessing invariance transport to every z , so H is a K -invariant element of $R([2+1])$, i.e. by Lemma 5.1 a homotopy of maps $W \rightarrow R$; it interpolates from $H_0 = s$ to H_1 , a cell prescribed by the box over an earlier stage, and H_1 is itself K -invariant (restrict the invariant H at $z = 1$). The induction hypothesis applied to H_1 finishes the step.

Limit stages add no cells. This exhausts all cases. $\square$

Remark 6.4. The proof uses no homology and no analysis of the homotopy type of R ; and it is uniformity that carries it. On a Reedy-type site one could try to run a similar induction with a naive fibrant replacement, but on $\square_{\text{DM}}$ the cell shapes of monomorphisms include quotients of representables — including Klein-quotient shapes, i.e. copies of the very object W — and an induction over such material would be circular. Uniform filling confines the fresh material to representable-shaped filler values (Lemmas 2.1 and 2.2), where the parity lemmas leave strict invariance no room: every invariant cell slides down its own filling cylinder onto earlier material, and at the bottom the strict quotient’s cells factor through explicitly contractible or explicitly null pieces. The contrast with the test side, where the category of elements absorbs the quotient into a Borel construction, is the entire content of the separation.

7. THE SEPARATION

Theorem 7.1. $\bar{\Phi} \in W_{\text{test}} \setminus W_{\text{type}}$. Consequently $W_{\text{type}} \subsetneq W_{\text{test}}$.

Proof. $\bar{\Phi} \in W_{\text{test}}$ is Theorem 4.1. Suppose $\bar{\Phi} \in W_{\text{type}}$. Let $u : J/K \rightarrow R$ and $u' : W \rightarrow R'$ be fibrant replacements. Then $R(\bar{\Phi}) : R \rightarrow R'$ is a weak equivalence between bifibrant objects, hence a homotopy equivalence; let $h : R' \rightarrow R$ be a homotopy inverse. The composite $g := h \circ u' : W \rightarrow R$ satisfies $R(\bar{\Phi}) \circ g \simeq u'$. By Theorem 6.3, g is homotopic to a constant, so u' is homotopic to a constant map. Since u' is a weak equivalence, this makes $\text{id}_{R'}$ factor through a point in the homotopy category, i.e. R' — and hence W — is contractible in the type structure, contradicting Corollary 2.3. $\square$

Corollary 7.2. *In the type structure, W admits no essential map to the type realization of $B\mathbb{Z}/2 \star B\mathbb{Z}/2$: writing R for a fibrant replacement of J/K (whose test type is the join), $[W, R] = \{*\}$. In particular the type homotopy type of W is neither contractible nor equivalent to the join: it is a De Morgan cubical set on which the two structures genuinely disagree.*

Corollary 7.3 (The isotropy trichotomy). *Combining with [14] and the companion results of this series: for isotropy quotients of De Morgan cubes, the type and test structures agree on the fixed-point-full family (both contractible) and on the free family (both classifying spaces), and disagree on*

the mixed family, where the Klein quotient of the square is a separating witness.

Remark 7.4 (What the type structure sees). The corollaries delimit, but do not compute, the type homotopy type of W . The strictification lemma reduces its homotopy and cohomology to computations with invariant cells of fibrant replacements of *other* objects, which the parity mechanism again constrains severely; a full determination of $\text{type}(W)$ — an explicit genuinely non-test homotopy type isolated in this setting — is an interesting open problem. We note only that all four candidate “twist” squares constructed from the loops of the two ends are null (they factor through one-dimensional segment material), so the type structure truncates the mixed quotient far more brutally than the test structure.

8. MACHINE CROSS-CHECKS

Every finite claim of this paper is verified by exhaustive computation, in scripts accompanying the artifact repository of this series:

<https://github.com/r-shibusawa/cubical-strict-j>

(release v1.14.0, commit e40ce4f9, archived at DOI 10.5281/zenodo.21834151; the scripts `scripts/collage_type*.py`). In particular: (i) the equivariance, self-duality and flow identities of Lemmas 3.2–3.3 are checked in the free De Morgan algebras on up to four generators, exactly, in the monotone-function representation; (ii) the classification of invariant cells (Proposition 5.4) is confirmed by the full census of the 168^3 level-two triples: 372 invariant classes (360 interior, 12 on the ends), the interior ones all with strictly invariant representatives — independently of the parity proofs given here; (iii) the collapse-locus closure properties (Lemma 4.2) are checked exhaustively at levels one and two; (iv) as negative controls, the census also verifies that no strict section of $\bar{\Phi}$ exists at either the cover or the quotient level, and that the four loop-twisted candidate squares are pairwise definitionally distinct before their common null-homotopy class is taken — the phenomena that motivated the parity analysis. A companion object-language development (the `LibJoinTC` module of the artifact’s Lean 4 kernel) realizes the strictly K -invariant twisted squares inside cubical type theory itself and machine-checks their invariance definitionally; it played a heuristic role and is not used by the proofs.

9. RELATED WORK AND OUTLOOK

That strict interval quotients break the correspondence between cubical models with reversals and spaces was established in [14] (completing a folklore expectation): both structures assign $K(\mathbb{Z}/2, 1)$ to $\square^1/\langle \neg \rangle$, whose realization is a point. The question separated here — whether the two *cubical* structures agree with each other — is of a different nature, and the answer localizes the disagreement precisely: it is not reversal as such (the free family

agrees), nor fixed points as such (the fixed-point-full family agrees), but the *mixture* of isotropy types, where the test structure builds a genuine Borel colimit while strict equivariance in the type structure is frozen by the parity lemmas.

The model structure comparison problem for cubical sites has been studied intensively in the Cartesian setting, where the equivariant refinement of [3] produces a type-theoretic model structure that *does* coincide with the test structure — making the De Morgan separation proved here a genuine dichotomy between sites rather than a defect of cubical methods. It is worth spelling out how the two situations differ. In the cartesian setting the obstruction addressed by [3] is already visible at the level of *objects*: symmetric quotients of cubes (the motivating example being the quotient of the square by the coordinate swap) fail to be contractible in the unrepaired type-theoretic structure, and the equivariant fibration condition is designed exactly so that such quotients contract, after which the repaired structure coincides with the test structure. On the De Morgan site the phenomenon is subtler: by [14], the Klein quotient W is non-contractible in *both* structures — indeed no isotropy quotient of the square separates the two localizers by object-level contractibility. The present paper therefore constructs a *map-level* witness, $\bar{\Phi} : J/K \rightarrow W$, whose test-side equivalence and type-side nullity expose a separation invisible to the object-level criterion; the responsible mechanism (the parity rigidity of strict K -invariance) has no cartesian counterpart, since it turns on the reversal. For the De Morgan site, [11] constructs the type structure and [4] analyses (relative) elegance; the expectation of non-coincidence goes back to [12]. Our separation shows that on the De Morgan site no Quillen equivalence between the two structures can exist along the identity functor; whether $\text{type}(\widehat{\square_{\text{DM}}})$ nonetheless presents spaces *abstractly* (via a different comparison) is exactly the question of computing localizations such as $\text{type}(W)$, which the nullity mechanism makes approachable.

Three further directions. First, the *scope of the parity mechanism*: Lemmas 5.2–5.3 are statements about $\mathbb{Z}/2$ -patterns in free De Morgan algebras; for other groups and dimensions the analogous character analysis should decide exactly which strict quotients separate the structures, refining the trichotomy of Corollary 7.3 into a complete classification. Second, the *Kleene and Boolean subvarieties*: the same questions on the sites of [13], where the fixed-line combinatorics differ. Third, *syntactic consequences*: the witness lives on the site of a proof assistant; the nullity theorem says definitional K -invariance forces homotopical triviality — a no-go complementary to those of [15], and a caution for internalizing strict quotients in cubical proof assistants.

Declarations.

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